GAS LAWS AND MOLE CONCEPT

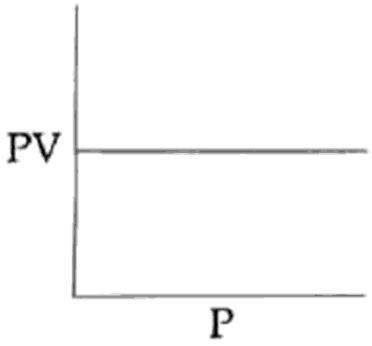
PYQ • 2021 - 2026

1. 2026 **3 MARKS**

Analyse the given graph drawn at constant temperature and answer the questions.

(a) State the gas law associated with this graph.

(b) The volume of a given mass of gas at a pressure 2 atm is 450 L. What will be the new volume if the pressure of this gas is increased 3 times ? (Temperature is constant)



2. 2026 **2 MARKS**

(A) The combustion equation for methane is given. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

(Atomic mass : C-12, H-1, O-16)

(a) Find the mass of oxygen required to react with 160 g of methane.

(b) Find the volume of CO_2 at STP produced by the complete combustion of 80 g of methane.

OR

(B) The volume of a sample of NH_3 gas at STP is 89.6 L. (Atomic mass : N-14, H-1)

(a) Calculate the mass of NH_3 sample.

(b) How many molecules are present in the sample ?

3. 2026 **1 MARK**

Find the relation and fill in the blanks.

The relationship between the number of molecules of a gas and its volume at constant temperature and pressure is given.

No. of molecules : x $\rightarrow$ Volume 20 L ; ..... $\rightarrow$ Volume 5 L

(a) 2x (b) x/4 (c) x/2 (d) x

4. 2025 **3 MARKS**

Volume and number of molecules of some gases at 27°C and 2 atm pressure are given in the table.

Nitrogen : 10 L, x molecules ; Carbon dioxide : (i) ___ L, 2x molecules ; Oxygen : 5 L, (ii) ___ molecules

(a) Complete the table.

(b) What will be the volume of carbon dioxide gas if the pressure is increased to 4 atm ? (Temperature and number of molecules remain constant)

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5. 2025 **2 MARKS**

Two gases Oxygen and Nitrogen at STP are given.

(a) Calculate the number of molecules in 64 g of Oxygen.

(b) Calculate the mass of Nitrogen having the same volume as that of 64 g of Oxygen.

(Hint : Atomic mass of Oxygen = 16, Nitrogen = 14)

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6. 2025 **1 MARK**

Find the total number of atoms in 2 GMM water (H_2O).

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7. 2024 **4 MARKS**

(a) What is the volume of 1 mol ammonia gas (NH_3) at STP ?

(b) Find the volume of 68 g ammonia at STP.

[Hint : Molecular mass of NH_3 = 17]

(c) Find the number of molecules present in 68 g ammonia gas.

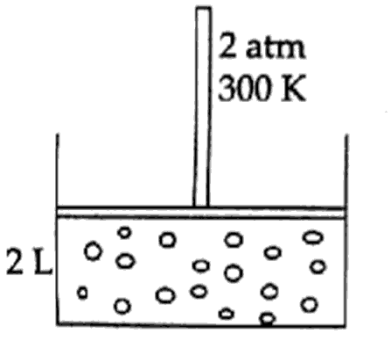
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8. 2024 **2 MARKS**

Analyse the picture and answer the following questions.

(a) What will be the pressure when the volume of this gas is increased to 4 L at 300 K ?

(b) Which is the gas law related to this ?



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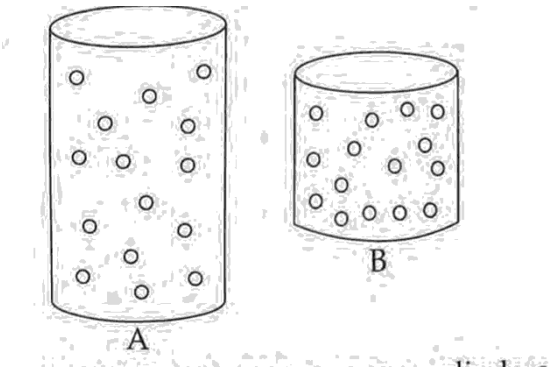
9. 2022 **4 MARKS**

A and B represent two gas cylinders. The gas in the cylinder A is completely transferred to cylinder B, keeping the temperature constant.

(a) Compare the gas pressure in cylinder A and cylinder B.

(b) Which gas law is related to this ?

(c) 10 L of a gas is kept in a cylinder at 2 atm pressure. Keeping the temperature constant, the gas is completely transferred to a 20 L cylinder. What is the new pressure of the gas ?



10. 2022 **2 MARKS**

Find the mass of 44.8 L of NH_3 kept at STP.

(Hint : Atomic mass N = 14, H = 1)

11. 2022 **1 MARK**

1 GMM of a substance contains _____ number of molecules.

12. 2021 **4 MARKS**

(a) What happens to the size of air bubble rising from the bottom of an aquarium ? What is the reason ?

(b) Which gas law is associated with this phenomenon ?

(c) State this law.

13. 2021 **4 MARKS**

From the given statements, write the statements applicable for gases.

(i) Energy of molecules is very high.

(ii) Attractive force between molecules is very high.

(iii) Distance between molecules is very large.

(iv) Freedom of movement of molecules is very less.

(v) As the collision of molecules is perfectly elastic in nature, there is no loss of energy.

(vi) When compared to total volume, the real volume of molecules is very less.

14. 2021 **3 MARKS**

The CO_2 gas kept at STP has volume 112 L.

[Hint : Molecular mass – 44]

(a) Find the number of moles of CO_2.

(b) Calculate the mass of 112 L CO_2.

(c) How many molecules of CO_2 are present in it ?

15. 2021 **1 MARK**

One gram atomic mass (1 GAM) of any element contains _____ atoms.